

EXPERIENCE

- ZS Associates, Bangalore, India – Advanced Data Science Associate Consultant

June 2025 – Present

 - Developed and deployed a production LLM-based classification pipeline to identify HCP engagement stage and therapy-area expertise, using historical interaction data and pharma activity signals processed via Python, SQL and Databricks, achieving $\approx 90\%$ classification accuracy and enabling structured insights into therapy area understanding
 - Designed, developed & deployed a personalized content navigator using RAG-enabled LLMs & Databricks to surface medical content based on prior HCP interactions, improving content relevance & reducing MSL preparation time.
- Ohio State University, Columbus, USA – Research Scholar

July 2024 – Feb 2025

 - Utilized Python, LLM, LoRA, NLP, along with Huggingface, Transformers, PyTorch and NumPy for efficient finetuning of LLMs in collaboration with Qualcomm, USA, employing layer selection strategies, layer offloading and novel parameter update approximation techniques resulting in 10% reduction in finetuning time and 60% GPU computational savings.
- Indian Institute of Technology, Bhilai, India – Researcher and TA

Aug 2023 – June 2024

 - Implemented a Federated Split Learning framework with caching and personalization, reducing communication and computation costs by 100 $\times$ under heterogeneous data conditions (March 2025 – May 2025)
 - Developed Big Data Technologies by configuring Hadoop, Hive, Spark and Kafka environments using Python, Docker, AWS and Kubernetes, achieving 100% functional setups (Aug 2023 – June 2024)
- Hewlett Packard Enterprise (HPE, R&D), Bangalore, India – Storage R&D Trainee

Jan 2023 – July 2023

 - Developed an end-to-end workflow in the HPE Storage framework using Python and libraries like Threading and ThreadPoolExecutor, resulting in more than 70% reduction in execution time (Offered FTE).

SKILLS

Languages and Tools	Python, SQL, AWS, Azure Cloud, DataBricks, Docker, Kubernetes, gRPC, Prometheus, Grafana, OpenAI API, Google Gemini API, PowerBI, GitHub Actions, Hadoop, Spark, Kafka, Storm, MongoDB, Qdrant VectorDB, Hive, Render, Android, Firebase, Java, R, C, Linux
DL/ML Techniques	LLM, NLP, Agentic AI, Generative AI, Data Science, Advanced RAG, Machine Learning, Deep Learning, Computer Vision, Split Learning, Federated Learning, Explainable AI, CNN, LSTM/RNN, ETL, Data Modeling and Mining, Supervised and Unsupervised learning techniques, Feature Engineering
Python Libraries	HuggingFace, PyTorch, Transformers, LangGraph, LangChain, Tensorflow, SKLearn, OpenCV, Pandas, Matplotlib, NumPy, LIME, FastAPI, Requests, BeautifulSoup

EDUCATION

Indian Institute of Technology, Bhilai – M.Tech in Data Science & AI (CGPA: 9.6/10 – 2nd Highest in Dept)	July 2023 – June 2025
PES University, Bengaluru – B.Tech in Computer Science and Engineering (CGPA: 9.0/10)	Aug 2019 – June 2023

PUBLICATIONS

- Adaptive Split Federated Learning – TechRxiv

Dec 2025

Built an adaptive split learning framework using gradient-norm-based inference, hybrid loss functions, activation caching, and personalization, achieving up to 100 $\times$ lower computation and communication cost and 29% higher test-time accuracy under heterogeneous, real-world data, making it robust and scalable across real-world domains.
- Benchmarking Multimodal AI for Telecom – ArXiv

Nov 2025

Curated a telecom benchmark suite, using LLMs and VLMs integrated with multimodal QA, RAG, and PCAP root cause analysis, resulting in enhanced model performance and actionable insights for adaptation to real-world telecom workflows.
- Automated and Interpretable Fake News Detection with XAI – Journal of Applied Security Research

May 2024

Designed a high-precision text and image-based news classification system, achieving an accuracy of over 92%.
- Enhancing Safety in Vehicles using Emotion Recognition with AI – IEEE Conference I2CT

April 2023

Engineered an in-vehicle system (Accuracy: 83% video, 78% audio) with music recommendations to stabilize driver emotion.

APPLIED AI PROJECTS AND INTERNSHIPS

- Agentic Marketing Mix (Feb 2026):

Built an Agentic workflow for automating marketing mix process for AI Hackathon in ZS
- Agentic AI System for Resume Analysis (Sept 2025):

Built an agentic AI-powered resume analyzer, using LangGraph and RAG pipeline with GitHub Actions for CI/CD to Azure, resulting in automated deployment and scalable evaluation with actionable feedback.
- AI Chatbot for Efficient Network Troubleshooting (Jan 2025):

Built an AI chatbot for efficient network troubleshooting, RAG-enhanced LLMs, resulting in optimized performance and scalability for real-time network issue resolution.
- CMT-SCTP Implementation (July 2022):

Muxable Labs Internship (2 months) involving contributing to open-source SCTP library.
- Observable AWS Application (Feb 2022):

HPE Internship (6 months) involving building a CI/CD pipeline using Python, Docker, Kubernetes, Prometheus and Grafana, achieving 100% real-time observability.